MINI PROJECT / MICROPROCESSORS AND MICROCONTROLLERS

8051 Hardware Lab

Real-time embedded monitoring on the AT89S52 - result sheet

ATHARVA (24BEL0082) · ADITYA (24BEL0087) · Electronics and Communication Engineering · 2025-26

Target AT89S52-24PU · 11.0592 MHz · 8 KB flash · 13 Keil C51 programs · 20 bench tests · INR 985 per board

What was built

A complete 8051 board: eight LEDs through a latch, an eight way DIP switch bank, a two digit seven segment display, a 16x2 HD44780 LCD, an LM35 through an ADC0804, a MAX232 serial link at 9600 8N1, a relay and a buzzer, and two interrupt buttons. The eight data lines of the LCD, the ADC and both output latches share one bidirectional bus on port 0, each device gated by its own strobe on port 2. **The peripherals require 54 I/O signals; the chip provides 32.** The 22 pin deficit is closed by architecture, with no port expander IC.

Headline results

Quantity	Designed value	Measured	Acceptance
I/O pins utilised	32 of 32		no headroom
Supply rail, U1 pin 40	5.00 V		4.85 - 5.15 V
UART baud error	0.00 %, TH1 = FDH		no framing errors
Display refresh	100 Hz per digit		95 - 105 Hz
Timer output, P3.4	1.00 kHz, FE34H		980 - 1020 Hz
One second tick	1.000 s, timer 0		+/- 0.2 s in 10 s
ADC step	7.8125 mV, Vref/2 = 1.000 V		reference 0.99 - 1.01 V
Temperature agreement	0.78 C per step		within 1.5 C, 25-60 C
Zone trip points	40.0 C and 70.0 C		+0 / - 0.5 C
Supply current	157 mA typ, 274 peak		under 200 mA
Largest program	13_selftest, 3.4 KB of 8 KB		fits AT89S52 flash

The Measured column is to be completed at the bench, in ink, during the review. Every tolerance above is the one stated in hardware/TESTING.md.

Constraints and how each was closed

ID	Constraint	Consequence if ignored	Mitigation implemented
C1	32 pins for 54 signals	peripherals dropped, or an expander IC added	one 8-bit bus on P0, per-device strobes on P2
C2	P0 is open drain, cannot source	the bus never reaches a valid logic 1	RN1, a 10 k x 9 SIP network, pulls the bus up
C3	output latches cannot be read back	a stale byte is latched into the wrong device	bus_shadow variable, interrupts gated per transfer
C4	P1.5-P1.7 shared with the ISP lines	in-circuit programming fails or is corrupted	1 k series resistors, SW6-SW8 documented OFF
C5	8-bit ADC, 5 V default reference	1.95 C per step, most of the range wasted	Vref/2 trimmed to 1.000 V, giving 0.78 C
C6	no hardware debounce on the buttons	one press registers as several interrupts	5 ms resampling, edge triggered INT0 and INT1

Operating behaviour

On power-up the board runs its program from flash with no host attached. The full system program monitors temperature against three zones - green below 40.0 C, yellow 40.0 to 69.9 C, red at 70.0 C and above - drives the zone LEDs, energises the relay at the trip point, sounds the buzzer, updates the LCD and display, and transmits one telemetry line per second at 9600 8N1. The emergency input removes the relay and forces the alarm state; the second input restores normal control. The relay is a low voltage demonstration load and is never connected to mains.

VERIFICATION / ALL THIRTEEN PROGRAMS

What each program demonstrates

#	Program	Result observed	Test	Verdict
01	led_blink	8 LEDs blink at 1 Hz; the board runs with no PC attached	T3	PASS / FAIL
02	led_patterns	sweep, then AA/55 alternation, then binary count	T3	PASS / FAIL
03	switch_led	each DIP switch lights its own LED, debounced	T4	PASS / FAIL
04	seg7_counter	two digit display counts 00 to 99, no ghosting	T5	PASS / FAIL
05	timer0_second	one count per second from the timer interrupt	T6	PASS / FAIL
06	square_wave	1.00 kHz on P3.4, 50 % duty, JP1 removed	T7	PASS / FAIL
07	ext_interrupt	PB1 alternates the pattern, one press one change	T8	PASS / FAIL
08	uart_echo	every byte typed on the terminal is echoed	T9	PASS / FAIL
09	uart_commands	HLP, LED FF, SEG NN, RLY 1, BUZ 1, TMP all respond	T10	PASS / FAIL
10	adc_7seg	LM35 temperature in degrees C on the display	T11	PASS / FAIL
11	adc_lcd	LCD shows temperature and a bar graph that tracks it	T12	PASS / FAIL
12	industrial_monitor	zones, relay, buzzer, LCD, emergency, telemetry	T13-17	PASS / FAIL
13	selftest	walks every subsystem, prints SELF TEST: PASS or FAIL	T1, T17	PASS / FAIL

All thirteen sources compile with 0 errors under Keil C51. The AT89S52 variants of the projects are the flashable ones; the AT89C51 has no ISP interface.

Bench test summary

#	Test	Acceptance	Result	#	Test	Acceptance	Result
T1	power on self test	PASS		T11	ADC and LM35	within 1.5 C	
T2	supply rails	5.00 +/-0.15 V		T12	LCD bar graph	monotonic, tracks	
T3	LED bank timing	20 half cycles / 10 s		T13	zone thresholds	40.0 / 70.0 C	
T4	DIP switch input	LED lights, no retrigger		T14	relay control	trips at 40.0 C	
T5	7 segment counter	50 +/-2 s for 00-99		T15	emergency stop	relay off, buzzer	
T6	one second timing	10 in 10.0 +/-0.2 s		T16	emergency reset	normal control	
T7	square wave	1.00 kHz +/-20 Hz		T17	telemetry rate	10 lines in 10 s	
T8	external interrupts	no double triggering		T18	current draw	under 200 mA	
T9	serial echo	byte exact, no framing		T19	regulator thermals	below 60 C	
T10	command shell	HLP, LED FF, RLY 1		T20	supply rejection	7-12 V in, 5 V out	

Demonstration sequence

1. Cold power-up with no host attached: program 01, eight LEDs at 1 Hz, showing the program is resident in flash. 2. Power-on self test, program 13, all subsystems reported on the terminal. 3. Full system, program 12: temperature raised and then lowered across the 40.0 C threshold, so the relay energises and releases. 4. Emergency input pressed and released, then normal control restored. 5. Telemetry observed at one line per second.

Stated limits

Area	Limit
Bus	shared P0 caps throughput to a few hundred kHz
ADC	one channel, 8-bit, about 114 us per sample
Memory	256 B RAM, no external data memory wired
Interrupts	two external sources, no priority nesting
Relay	demonstration load only, never mains connected

Reviewed by _____ **Signature / date** _____

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